

Suggested problems: 9, 11, 13, 14

Problem 1: (Example 2.1.9) Two boxes contain long bolts and short bolts. Suppose that one box contains 60 long bolts and 40 short bolts, and that the other box contains 10 long bolts and 20 short bolts. Suppose also that one box is selected at random and a bolt is then selected at random from that box. What is the probability that the bolt is long?

Problem 2: (Problem 2.1.13) A box contains three coins with a head on each side, four coins with a tail on each side, and two fair coins. If one of these nine coins is selected at random and tossed once, what is the probability that a head will be obtained?